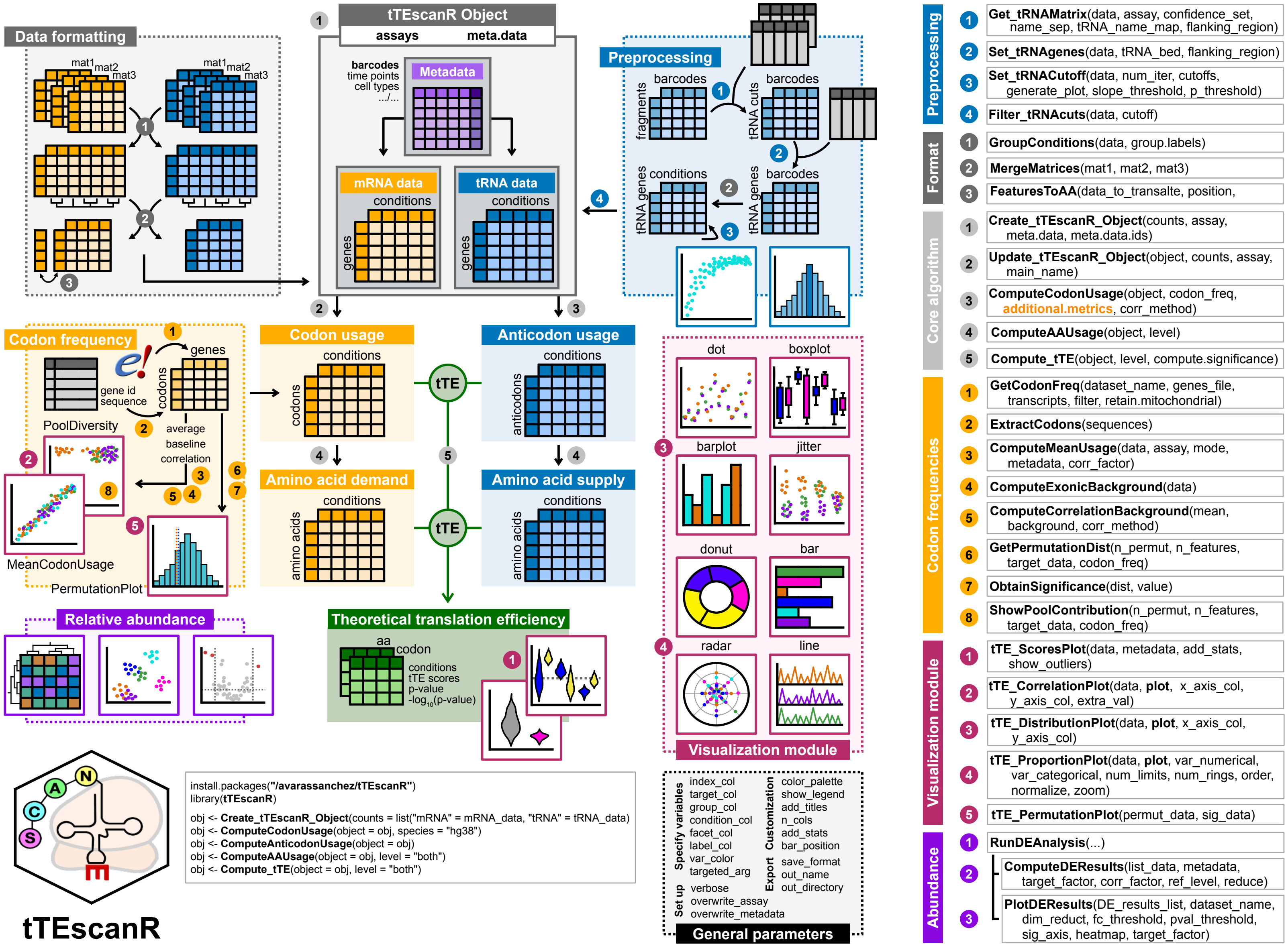

Data formatting
mat1 mat2 mat3
tTEscanR Object
assays
meta.data
barcodes
time points
cell types
.../...
Metadata
mRNA data
tRNA data
conditions
genes
Preprocessing
barcodes
fragments
tRNA cuts
conditions
tRNA genes
Codon frequency
genes
gene id
sequence
codons
PoolDiversity
average
baseline
correlation
MeanCodonUsage
PermutationPlot
Codon usage
conditions
codons
Anticodon usage
anticodons
tTE
Amino acid demand
amino acids
Amino acid supply
Theoretical translation efficiency
aa
codon
conditions
tTE scores
p-value
-log10(p-value)
Relative abundance
dot
boxplot
barplot
jitter
donut
bar
radar
line
Visualization module
Specify variables
index_col
target_col
group_col
condition_col
facet_col
label_col
var_color
targeted_arg
Customization
color_palette
show_legend
add_titles
n_cols
add_stats
bar_position
Export
save_format
out_name
out_directory
Set up
verbose
overwrite_assay
overwrite_metadata
General parameters
install.packages("/avarassanchez/tTEscanR")
library(tTEscanR)
obj <- Create_tTEscanR_Object(counts = list("mRNA" = mRNA_data, "tRNA" = tRNA_data))
obj <- ComputeCodonUsage(object = obj, species = "hg38")
obj <- ComputeAnticodonUsage(object = obj)
obj <- ComputeAAUsage(object = obj, level = "both")
obj <- Compute_tTE(object = obj, level = "both")
tTEscanR
Preprocessing
1 Get_tRNAMatrix(data, assay, confidence_set, name_sep, tRNA_name_map, flanking_region)
2 Set_tRNAgenes(data, tRNA_bed, flanking_region)
3 Set_tRNACutoff(data, num_iter, cutoffs, generate_plot, slope_threshold, p_threshold)
4 Filter_tRNAcuts(data, cutoff)
Format
1 GroupConditions(data, group.labels)
2 MergeMatrices(mat1, mat2, mat3)
3 FeaturesToAA(data_to_transalte, position,
Core algorithm
1 Create_tTEscanR_Object(counts, assay, meta.data, meta.data.ids)
2 Update_tTEscanR_Object(object, counts, assay, main_name)
3 ComputeCodonUsage(object, codon_freq, additional.metrics, corr_method)
4 ComputeAAUsage(object, level)
5 Compute_tTE(object, level, compute.significance)
Codon frequencies
1 GetCodonFreq(dataset_name, genes_file, transcripts, filter, retain.mitochondrial)
2 ExtractCodons(sequences)
3 ComputeMeanUsage(data, assay, mode, metadata, corr_factor)
4 ComputeExonicBackground(data)
5 ComputeCorrelationBackground(mean, background, corr_method)
6 GetPermutationDist(n_permut, n_features, target_data, codon_freq)
7 ObtainSignificance(dist, value)
8 ShowPoolContribution(n_permut, n_features, target_data, codon_freq)
Visualization module
1 tTE_ScoresPlot(data, metadata, add_stats, show_outliers)
2 tTE_CorrelationPlot(data, plot, x_axis_col, y_axis_col, extra_val)
3 tTE_DistributionPlot(data, plot, x_axis_col, y_axis_col)
4 tTE_ProportionPlot(data, plot, var_numerical, var_categorical, num_limits, num_rings, order, normalize, zoom)
5 tTE_PermutationPlot(permut_data, sig_data)
Abundance
1 RunDEAnalysis(...)
2 ComputeDEResults(list_data, metadata, target_factor, corr_factor, ref_level, reduce)
3 PlotDEResults(DE_results_list, dataset_name, dim_reduct, fc_threshold, pval_threshold, sig_axis, heatmap, target_factor)